SAMPLE

Cosmic Explorers: Astronaut Training

From Selection to Spaceflight

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Free sample: Introduction & Chapter 1

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Introduction

The room looks unremarkable: a pool of water, a rack of clothing, a flight surgeon with a clipboard. What happens in it is anything but. Over there, a person in a white spacesuit is being lowered by crane into the water, to rehearse, step by step, a repair that will one day be performed four hundred kilometers above the Earth, at seventeen thousand kilometers an hour, where a mistake cannot be undone. Over here, another candidate hangs in a harness, learning to work a wrench in a world where the wrench works you back. Everything in the building is aimed at a single purpose: to make the strangest environment our species has ever entered feel, by launch day, almost familiar.

That is astronaut training, and this book is about it. The word astronaut suggests a person who rides a rocket, and the public image stops there, at the launch and the floating and the view. But the ride is the smallest part of the job. The years before it are where the profession actually lives: the selection that chooses a handful of applicants from tens of thousands, the classroom work on orbital mechanics and systems engineering, the pools and centrifuges and simulators that teach the body and the reflexes, the survival schools that prepare for the day the machine fails, the psychology and the crew dynamics that keep a sealed group of six sane for six months, and the unglamorous hours of procedure rehearsal that turn a laboratory falling around the planet into a workplace.

Training is long, and its structure has a rhythm worth stating up front. Candidates arrive as accomplished people, pilots, scientists, engineers, physicians, and then spend roughly two years in basic training before they are even called astronauts. Advanced training follows, on the systems and skills of the trade, and then mission-specific training, the focused years in which a crew rehearses its particular flight, which for a long-duration mission runs about eighteen months more. The training does not stop at launch: onboard refresher sessions and proficiency maintenance continue for the whole mission, and rehabilitation, the training of the body to be an Earthling again, follows the landing.

The scale of the effort is the honest theme of this book.

It is also worth saying what training is not, because the public conversation about astronauts runs on the wrong fuel. The profession is not about physical courage, or at least not primarily: the medical and operational filters select for competence, and the training installs the procedures that courage would only improvise. Nor is it about genius, exactly: the astronaut corps contains plenty of brilliance, but the pipeline's real product is reliability, the trained ability to do the correct thing, at the correct time, for months, without a net. The chapters that follow present the most elaborate system our species has built for making a small number of ordinary-but-extraordinary people dependable in the least dependable environment we have ever worked in, and the system's size, the pools and the jets and the decades, is the honest measure of how hard that is. A space mission is the most expensive machine operation humans perform, in money and in risk, and training is how the risk is paid down. Every hour of rehearsal is an hour of failure removed from orbit. Every pool session, every simulation, every emergency drill is a bet that the real emergency, if it comes, will find people who have already been there.

The chapters that follow walk the whole pipeline. They begin with selection, the remarkable sorting operation that decides who gets to train. They cover the body: what weightlessness does to bones, muscles, eyes, and the inner ear, and how training and countermeasures fight back. They follow the candidate through the great machines of the trade, the centrifuge, the neutral buoyancy pool, the simulators, the survival schools, and the virtual realities, and through the classroom disciplines, systems, robotics, procedures, and the sciences of the mission itself. They look at the mind: the psychology of isolation, the crew as a team, the analog environments where the lessons are tested on Earth. And they look forward, at the training being invented right now for the return to the Moon and the first voyages to Mars, for which no one alive has a template.

One promise before the tour begins, the same one this series always makes: real science and real practice, honestly explained, with the wonder left in. The people in the pool and the centrifuge are doing one of the most demanding jobs ever defined, and the machinery they use to prepare is a story worth the telling.

Chapter 01: Selection: The First Cut

Before the first day of training, there is the sorting. Astronaut candidate selection is one of the most competitive processes on Earth, and understanding it explains much of what follows, because the training system is built for people who already survived a filter most people could not.

The numbers set the tone. When a major space agency opens a candidate class, the applications arrive in the tens of thousands, and the selection funnel ends with a class of a handful to a dozen. The filter is not a single test but a long sequence: eligibility screens, medical examinations, interviews, team exercises, and background checks, spread over a year or more. The survivors of that funnel are the people the training system receives, and training is designed with that in mind: it can assume, at entry, an unusual combination of health, competence, and stability.

Eligibility itself is a high bar, and it has widened over the decades.

The widening had a turning point worth a paragraph, because it explains the modern profession's shape. The shuttle era brought the payload specialist, the scientist flying for a specific mission, and then the class that changed everything, the era's teacher and the professional mission specialists, and the curriculum had to absorb people whose flight experience was a classroom. The training system discovered that its real prerequisites were not jet hours but the learned habits of high performance: disciplined study, procedural memory, composure under observation, and the program began teaching the flying from scratch, a bet the selection had made on temperament and the training made good. The bet is the modern profession's foundation: the station era's crews are mostly career scientists and engineers who have been built into pilots, divers, mechanics, and medics by the pipeline, and the pipeline's confidence that it can build those skills into a doctor or a geologist is the same confidence that once flew test pilots to the Moon. In the early era, astronauts were drawn almost entirely from military test pilots, and the selection criteria were a fighter pilot's: thousands of hours of jet time, an engineering degree, perfect physiology. As the profession changed from flying to living and working in space, the criteria widened to absorb scientists, physicians, engineers, and educators, and the modern class mix reflects it: the pilot's slot remains, but most members of a modern candidate class are mission and payload specialists, chosen for advanced degrees and operational aptitude rather than cockpit hours. The recipe varies by agency, but the shape is universal: demonstrated excellence in a technical field, evidence of hands-on skill, and the interpersonal makeup to be a good crewmate in a sealed can for months.

The medical examination merits its own paragraph, because it is unlike any ordinary physical.

The screenings that most surprise candidates are the functional ones. The vestibular battery puts the inner ear through its paces, caloric tests and rotation, precisely because the first days of a mission will punish any weakness there; the cardiovascular testing includes tilt tables and lower-body negative pressure, the devices that reproduce the fluid shifts of weightlessness on the ground, so the examiners can watch the body respond while the physicians still have the option of a disqualification. The eye standards, the dental standard, the hearing standard, all have the same logic: a medical emergency in orbit is a mission emergency, and the cheapest medical care is the condition that was never aboard. The exam, in other words, is not a health check but a flight-readiness check, and the candidates who pass it have been pre-flown, in effect, before their training begins. Candidates undergo a battery lasting days: detailed imaging, vestibular and cardiovascular testing, vision and hearing standards, dental checks, psychological screening, and a deep history review, because the agency is not looking merely for health but for robustness, a body that can absorb the insults of spaceflight without a crisis. Standards have eased somewhat as medicine learned which conditions actually matter in orbit, but the exam remains a funnel that removes more candidates than any interview.

The interviews and team exercises probe the rest. Selection boards look for composure under interrogation, the ability to admit mistakes, the temperament to be a follower on one mission and a leader on the next, and the specific, unglamorous quality of being easy to live with under stress. Group exercises put candidates through realistic operational problems while evaluators watch who listens, who shares credit, who cracks. The astronaut corps is a permanent small team, and selection is, in large part, choosing colleagues for a workplace where no one can go home at the end of the day.

The result, on the first day of basic training, is a candidate class that is, by construction, the output of one of the world's most aggressive filters.

The selection is also an international patchwork, and the geography is worth knowing. The American program selects through a periodic open call, years apart, that any qualified citizen may answer, and the class that emerges is famously mixed, physicists and submariners, surgeons and test pilots, in a single cohort. The Russian program draws largely from the military and the aviation world, with its own Korolev-era traditions. The European, Canadian, and Japanese agencies each run their own campaigns, smaller and less frequent, on similar principles, and the international partner astronauts then cross-train in each other's facilities, which means a candidate class anywhere in the partnership is likely to contain a future crewmate from another continent, learning the same procedures in another language. The filters differ in detail; the output profile, technical excellence plus operational temperament

plus physical robustness, is the same everywhere, because the machine the candidates are being trained for is the same. The years that follow are not designed to find talent, but to build it: to take people who are already extraordinary and make them, through two years of structured immersion, into the specialists the mission needs. The next chapter describes the machine that does it.